

AI AI &

AI

2025-11-20

Table of contents

1	AI &	3
2	(AI & Machine Learning Fundamentals)	4
2.1		4
2.2		4
2.3		5
2.4		5
3	AI (AI Applications & Generative AI)	6
3.1	AI	6
3.2	AI	6
3.3	AI	7
4	(Data Science & Big Data)	8
4.1		8
4.2		8
5	AI (AI Ethics, Governance & Risks)	10
5.1	AI	10
5.2	AI	10
6	AI (AI Programming)	11
6.1	Python	11

1 AI AI &

iPAS AI /

AI

AI

AI

Python

2 (AI & Machine Learning Fundamentals)

2.1

- (L111)
 - AI (L11101)
 - *
 - * AI
 - (ANI) (AlphaGo, Siri)
 - (AGI)
 - (ASI) AI ()

2.2

- (L113)
 - (L11301)
 - * (Machine Learning)
 - * AI
 - (L11302)
 - *
 - *
 - (Linear Regression) (Logistic Regression)
 - (Decision Tree)
 - (Random Forest)
 - (SVM)
 - (Naive Bayes)
 - K- (KNN)
 - K- (K-Means)
- (L231)

- / (L23101)
- (L23102)
- (L23103)
- * ()

2.3

- (L232)
 - (L23201)
 - (L23202)
 - (L23203)
 - *
 - *
 - (CNN) (ResNet, VGG)
 - (RNN/LSTM/GRU)
 - Transformer (BERT, GPT)
- (L332)
 - (Accuracy) (Recall)
 - - * Accuracy, Precision, Recall, F1-Score, ROC/AUC
 - * Bagging (Random Forest), Boosting (XGBoost), Stacking
 - * L1 (Lasso), L2 (Ridge), Dropout, Batch Normalization

2.4

- (L233)
 - (L23301)
 - * One-hot Encoding, / ,
 - (L23302)
 - (L23303)
 - * (Cross-validation), (Confusion Matrix)
 - (L22304)
 - * (Grid Search)

3 AI AI (AI Applications & Generative AI)

3.1 AI

- AI AI (L114)
 - AI AI (L11401)
 - * AI (Discriminative AI) AI (Generative AI)
 - * (/ vs)
 - AI AI (L11402)
 - *

3.2 AI

- AI (L12)
 - No Code / Low Code (L121)
 - * No Code / Low Code (L12101)
 - No Code () Low Code ()
 - * No Code / Low Code (L12102)
 - AI
 - Zapier, Make, Coze, Dify
 - AI (L122)
 - * AI (L12201)
 -
 - ChatGPT, Claude, Midjourney, Stable Diffusion, GitHub Copilot
 - * AI (L12202)
 - Prompt Engineering (Zero-shot, Few-shot, CoT)
 - AI (L123)
 - * AI (L12301)
 - * AI (L12302)

- * AI (L12303)
- AI (L343)
 - AI

3.3 AI

- (L21)
 - AI (L211)
 - * (L21101)
 - * (L21102)
 - * AI (L21103)
 - * (L21104)
 - * AI
 - *
 - GAN
 - Diffusion Model
 - LLM (Transformer)
 - AI (L212)
 - * AI (L21201)
 - * AI (L21202)
 - * AI (L21203)
 - AI (L213)
 - * (L21301)
 - * AI (L21302)
 - * MLOps, Edge AI, Cloud AI, API
- (L342)
 - (Bag-of-Words)
 -
 - NLP TF-IDF, Word2Vec, BERT, GPT
 - CV YOLO (), Segmentation ()

4 (Data Science & Big Data)

4.1

- (L112)
 - (L11201)
 - *
 - (L11202)
 - *
 - * AI
 - (L11203)
 - *
- (L331)
 - (Pie Chart)
 -
 -
 - * (Bar Chart)
 - * (Line Chart)
 - * (Histogram) ()
 - * (Scatter Plot)
 - * (Box Plot)
 - * (Heatmap)

4.2

- (L22)
 - (L221)
 - * (L22101)
 - * (L22102)
 - * (L22103)
 - *

- **(L222)**
 - * **(L22201)**
 - * **(L22202)**
 - * **(L22203)**
 - *
 - * Hadoop (MapReduce), Spark ()
 - * **NoSQL** MongoDB (), Cassandra (), Redis ()
 - * Kafka ()
- **(L223)**
 - * **(L22301)**
 - * **(L22302)**
 - * **(L22303)**
 - *
- **(L224)**
 - * **(L22401)**
 - * **AI (L22402)**
 - * **AI (L22403)**
 - * **(L22404)**
 - * **AI**

5 AI (AI Ethics, Governance & Risks)

5.1 AI

- AI (L11102)
 - (Privacy) (Security) AI (Ethics)
 - AI
- (L234)
 - (L23401)
 - (L23402)
- AI (L341)
 - AI (AI Hallucination)
 - AI
 - AI
 - * (Bias)
 - * Deepfake AI
 - * (Adversarial Attacks) AI
 - * (Model Drift)

5.2 AI

- AI (L31)
 - AI (L311)
 - * (Data Anonymization) (User Consent Management)
 - *
 - AI (L312)
 - *
 - *
 - * AI

6 AI (AI Programming)

6.1 Python

- AI Python (L32)

- (L321)

- *
* 3_Pig
* print(5, 10, sep=',') 5,10

- (L322)

- * //()
* ** (), % ()
*
· while...else else
· for (List, String)
· break continue

- (L323)

- * List (), Tuple (), Dictionary ()
* Dictionary (key-value) Tuple
* (Set)